

ESAT PRACTICE TEST SERIES

PHYSICS + CHEMISTRY

LEVEL 0 • EASY

Physics + Chemistry

Mathematics 1 + Physics + Chemistry

Foundation-focused ESAT practice with more direct setup and shorter solution chains.

MODULE

Mathematics 1

MODULE

Physics

MODULE

Chemistry

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions • three verified module keys

Physics + Chemistry • Level 0 Easy

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Physics + Chemistry. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 1 · preserved Mathematics 1 module

Q1 G	Q2 C	Q3 B	Q4 B	Q5 E
Q6 G	Q7 D	Q8 C	Q9 B	Q10 C
Q11 A	Q12 D	Q13 B	Q14 A	Q15 D
Q16 C	Q17 B	Q18 D	Q19 C	Q20 E
Q21 E	Q22 C	Q23 E	Q24 D	Q25 E
Q26 C	Q27 C			

Physics

Engineering Mock Test 1 · preserved Physics module

Q1 F	Q2 A	Q3 D	Q4 E	Q5 G
Q6 C	Q7 D	Q8 E	Q9 D	Q10 D
Q11 F	Q12 B	Q13 C	Q14 G	Q15 D
Q16 E	Q17 B	Q18 C	Q19 G	Q20 D
Q21 E	Q22 G	Q23 H	Q24 C	Q25 D
Q26 C	Q27 B			

Chemistry

Preserved Chemistry Easy source set

Q1 B	Q2 D	Q3 C	Q4 D	Q5 D
Q6 D	Q7 F	Q8 C	Q9 B	Q10 H
Q11 C	Q12 A	Q13 D	Q14 C	Q15 B
Q16 A	Q17 F	Q18 B	Q19 C	Q20 B
Q21 C	Q22 D	Q23 B	Q24 C	Q25 C
Q26 E	Q27 C			

Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 1 - Mathematics 1

Answer **G****QUESTION 1**

If the 5-digit number $1X2X3$ is divisible by 9, what is X ?

A 0**B** 1**C** 2**D** 3**E** 4**F** 5**G** 6**WORKED SOLUTION**

The digit sum is $1 + X + 2 + X + 3 = 6 + 2X$. For divisibility by 9, $6 + 2X$ must be a multiple of 9. From the choices, $X = 6$ gives 18.

QUESTION 2

If we define $a \star b = a^2 - b^2$, what is $5 \star 4$?

A 1

B 7

C 9

D 16

E 25

F 41

WORKED SOLUTION

Apply the definition directly: $5 \star 4 = 5^2 - 4^2 = 25 - 16 = 9$.

QUESTION 3

Jack tosses two fair dice. What is the probability that the sum of the two values is no less than 9?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number of all possible cases}}.$$

A $\frac{1}{4}$

B $\frac{5}{18}$

C $\frac{1}{3}$

D $\frac{5}{12}$

E $\frac{2}{3}$

WORKED SOLUTION

For two dice, sums at least 9 are 9, 10, 11, 12, giving $4 + 3 + 2 + 1 = 10$ outcomes. So the probability is $10/36 = 5/18$.

M1-04

Mock Test 1 - Mathematics 1

Answer **B**

QUESTION 4

How many 4-digit integers $\overline{abcd}$ satisfy that

- 1 The thousands digit a and the tens digit c are even numbers;
- 2 The hundreds digit b and the unit digit d are odd numbers.
- 3 This number is divisible by 5?

A 80

B 100

C 125

D 250

E 500

WORKED SOLUTION

Since the number is divisible by 5, the final digit must be 5 because d is odd.

Then $a \in \{2, 4, 6, 8\}$, b has 5 choices, and c has 5 choices, giving

$$4 \cdot 5 \cdot 5 \cdot 1 = 100.$$

M1-05

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 5

If $\frac{m}{10} = \frac{6}{n} = \frac{2}{5}$, what is $m + n$?

A 10

B 13

C 14

D 16

E 19

WORKED SOLUTION

From $\frac{m}{10} = \frac{2}{5}$, $m = 4$. From $\frac{6}{n} = \frac{2}{5}$, $30 = 2n$, so $n = 15$. Hence $m + n = 19$.

QUESTION 6

Nancy has 100 marbles. She gives 20% of them to her friend Pedro. Then Nancy gives 25% of what is left to another friend, Ben. Finally, Nancy gives 10% of what is now left in the bag to her brother Jimmy. What percentage of her original bag of marbles does Nancy have left for herself?

A 20

B 30

C $33\frac{1}{3}$

D 38

E 45

F 50

G 54

H 60

WORKED SOLUTION

Track what remains: after 20% is given away, 80 remain; after giving 25% of 80, 60 remain; after giving 10% of 60, 54 remain.

QUESTION 7

What is the sum of all positive 2-digit integers that have a remainder of 1 when divided by 9 and a remainder of 2 when divided by 5?

A 29

B 56

C 84

D 119

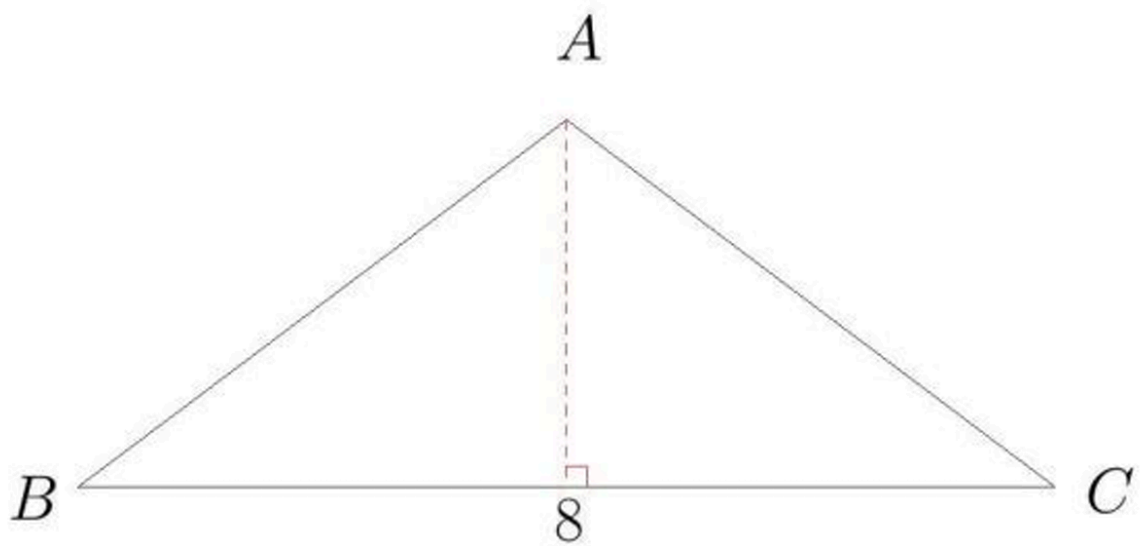
E 164

WORKED SOLUTION

The numbers satisfying both conditions are 37 and 82: each is 1 more than a multiple of 9 and 2 more than a multiple of 5. Their sum is 119.

QUESTION 8

$\triangle ABC$ is an isosceles triangle. $AB = AC$. If $BC = 8$ and the area of $\triangle ABC$ is 12, what is the perimeter of $\triangle ABC$?



A 12

B 16

C 18

D 19

E 20

WORKED SOLUTION

The height is h where $\frac{1}{2} \cdot 8 \cdot h = 12$, so $h = 3$. The altitude bisects the base, giving half-base 4, so each equal side is $\sqrt{3^2 + 4^2} = 5$. Perimeter $= 8 + 5 + 5 = 18$.

QUESTION 9

What is the smaller angle between the hour hand and minute hand of a clock at 5:50 pm?

A 100°

B 125°

C 150°

D 155°

E 165°

WORKED SOLUTION

At 5:50, the minute hand is at 300° . The hour hand is at $5 \cdot 30 + 50 \cdot 0.5 = 175^\circ$. The smaller angle is $300 - 175 = 125^\circ$.

QUESTION 10

The average age of 5 people in a room is 40 years. A 36-year-old person leaves the room. What is the average age of the four remaining people?

A 39

B 40

C 41

D 43

E 45

WORKED SOLUTION

The original total age is $5 \cdot 40 = 200$. After the 36-year-old leaves, the total is 164, so the new average is $164/4 = 41$.

M1-11

Mock Test 1 - Mathematics 1

Answer **A**

QUESTION 11

In how many ways can we write 127 as the sum of two prime numbers?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

An odd number written as a sum of two primes must use 2 plus an odd prime. Here $127 = 2 + 125$, but 125 is not prime, so there are no ways.

QUESTION 12

If for some positive integer n ,

$$\left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) \left(1 + \frac{1}{4}\right) \cdots \left(1 + \frac{1}{n}\right) = 50,$$

what is n ?

A 20

B 39

C 65

D 99

E 100

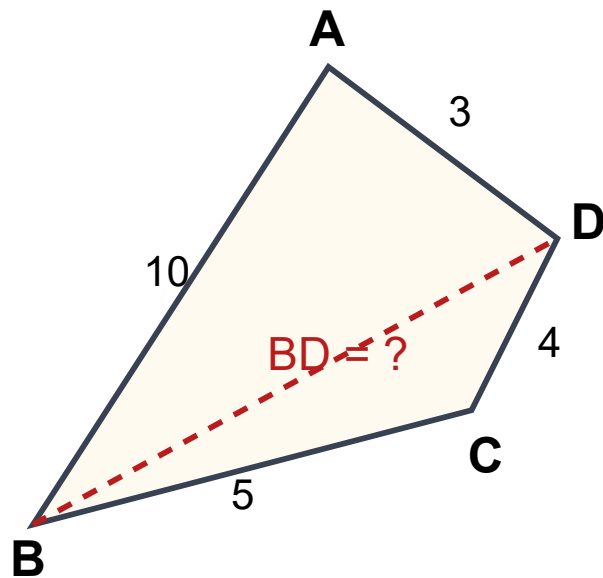
WORKED SOLUTION

Each factor is $1 + \frac{1}{k} = \frac{(k+1)}{k}$. The product telescopes:

$$\frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = \frac{n+1}{2} = 50. \text{ Thus } n = 99.$$

QUESTION 13

In quadrilateral $ABCD$, $AB = 10$, $BC = 5$, $CD = 4$ and $AD = 3$. If the length of BD is an integer, what is BD ?



A 7

B 8

C 9

D 10

E there is not enough information

WORKED SOLUTION

Use the triangle inequality in the two triangles that share BD . For $\triangle ABD$, $|10 - 3| < BD < 10 + 3$, so $7 < BD < 13$. For $\triangle BCD$, $|5 - 4| < BD < 5 + 4$, so $1 < BD < 9$. The only integer satisfying both is $BD = 8$.

QUESTION 14

What is the unit digit of 7^{2020} ? Here

$$7^n = \underbrace{7 \times 7 \times 7 \times \dots \times 7}_n$$

(Hint: check unit digits of 7 , 7^2 , 7^3 , ... and find the pattern.)

A 1**B** 3**C** 5**D** 7**E** 9**WORKED SOLUTION**

The units digits of powers of 7 cycle as 7, 9, 3, 1. Since 2020 is divisible by 4, the units digit is the fourth term of the cycle, 1.

QUESTION 15

Here x , y , z and u represent four different numbers.

$$\begin{array}{r} x \quad y \quad x \quad y \\ + z \quad x \quad - z \quad x \\ \hline u \quad x \quad \quad x \end{array}$$

$$x + y + z + u =$$

A 12

B 14

C 16

D 18

E 20

WORKED SOLUTION

From the subtraction, $(10x + y) - (10z + x) = x$, so $8x + y = 10z$. From the addition, $(10x + y) + (10z + x) = 10u + x$, so $10x + y + 10z = 10u$. The distinct digit solution is $(x, y, z, u) = (5, 0, 4, 9)$, giving 18.

M1-16

Mock Test 1 - Mathematics 1

Answer C

QUESTION 16

Billy, a seasonal worker in the town of Cowra, collected 8 buckets of cherries on his first day. Each day after that he increased the number of buckets he picked by 2 buckets per day. According to this pattern, how many buckets will he collect in the first 20 days?

A 308

B 500

C 540

D 580

E 600

WORKED SOLUTION

The buckets form an arithmetic sequence: 8, 10, 12, The 20th term is $8 + 19 \cdot 2 = 46$. Sum = $\frac{20}{2}(8 + 46) = 540$.

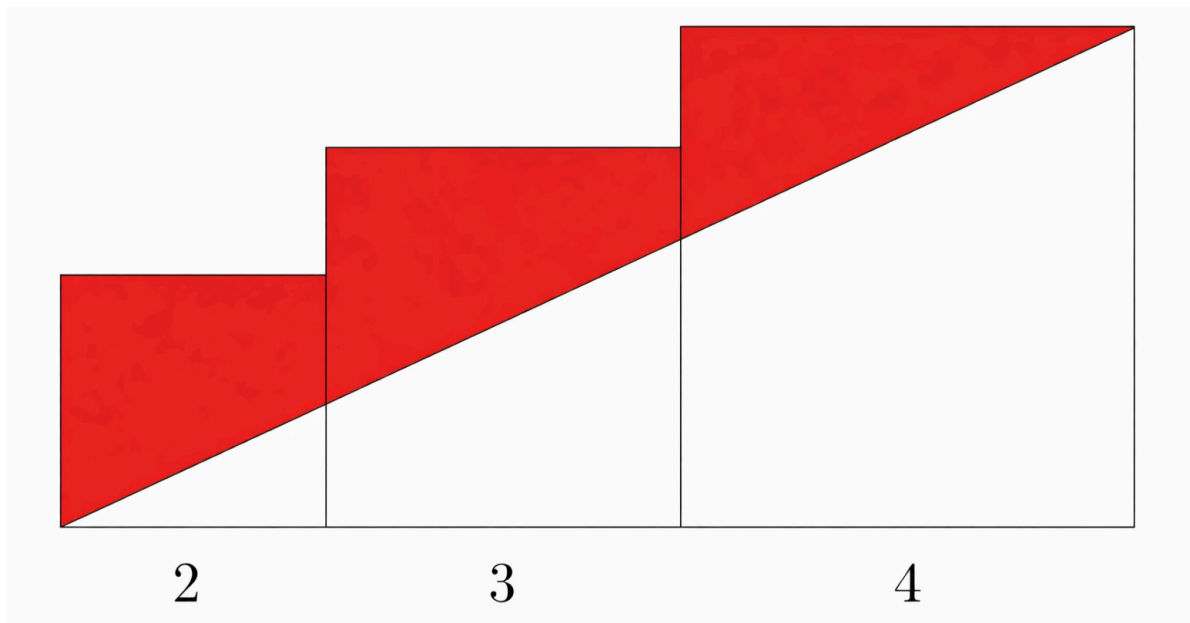
M1-17

Mock Test 1 - Mathematics 1

Answer **B**

QUESTION 17

Three squares are lined up horizontally as shown. Their side lengths are 2, 3 and 4 respectively. A straight line is drawn from the top right corner of the largest square to the bottom left hand corner of the smallest square. What is the area of the shaded region?



A 10

B 11

C 13

D 16

E 18

WORKED SOLUTION

The total area of the three squares is $2^2 + 3^2 + 4^2 = 29$. The line cuts off one large triangle of base 9 and height 4, area 18. Shaded area = $29 - 18 = 11$.

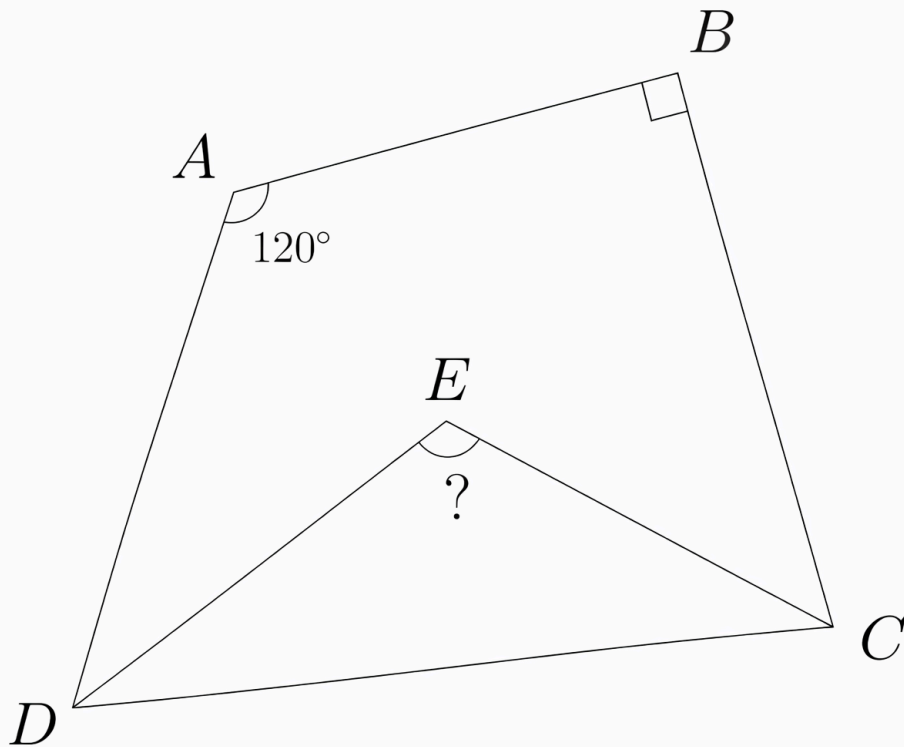
M1-18

Mock Test 1 - Mathematics 1

Answer D

QUESTION 18

In quadrilateral $ABCD$, $\angle A = 120^\circ$ and $\angle B = 90^\circ$. ED and EC bisect $\angle ADC$ and $\angle BCD$ respectively. What is $\angle E$?



A 90

B 95

C 100

D 105

E 110

WORKED SOLUTION

In the quadrilateral, $\angle C + \angle D = 360^\circ - 120^\circ - 90^\circ = 150^\circ$. Since EC and ED bisect those angles, triangle DCE has angles at C, D summing to 75° . Therefore $\angle E = 180^\circ - 75^\circ = 105^\circ$.

M1-19

Mock Test 1 - Mathematics 1

Answer C

QUESTION 19

In a mathematics contest with 10 problems, a student gains 6 points for a correct answer and loses 1 point for an incorrect answer. If Olivia answered every problem and her score was 32, how many correct answers did she have?

A 4

B 5

C 6

D 7

E 8

WORKED SOLUTION

Let c be the number correct. Then incorrect answers are $10 - c$, so
 $6c - (10 - c) = 32$. Hence $7c = 42$ and $c = 6$.

M1-20

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 20

Emily and Owen are going to a library that is 1 mile from their house. They leave home simultaneously. Emily rides her bicycle to the library at a constant speed of 12 miles per hour. Owen walks to the library at a constant speed of 5 miles per hour. How many minutes before Owen does Emily arrive?

A 3

B 4

C 5

D 6

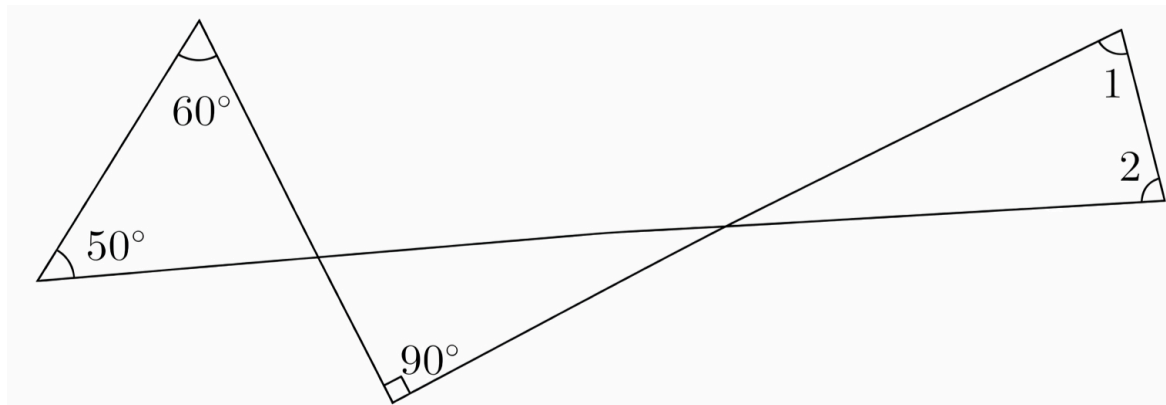
E 7

WORKED SOLUTION

Emily takes $1/12$ hour = 5 minutes. Owen takes $1/5$ hour = 12 minutes. Emily arrives $12 - 5 = 7$ minutes earlier.

QUESTION 21

In the following picture, if $\angle 1 = \angle 2$, what is $\angle 1$?



A 60°

B 65°

C 70°

D 75°

E 80°

WORKED SOLUTION

The left triangle gives the angle at the first intersection as

$180^\circ - 50^\circ - 60^\circ = 70^\circ$. The next triangle has a right angle, so the angle at the second intersection is 20° . The final small triangle has angles x , x , 20° , so $2x = 160^\circ$, giving $x = 80^\circ$.

QUESTION 22

Jack, a zoo keeper, is going to give N peaches to a group of monkeys. If he gives 10 peaches to each monkey, two monkeys will receive no peach. Or, the peaches are exactly enough to give 8 to each monkey. What is the sum of digits of N ?

A 5

B 6

C 8

D 9

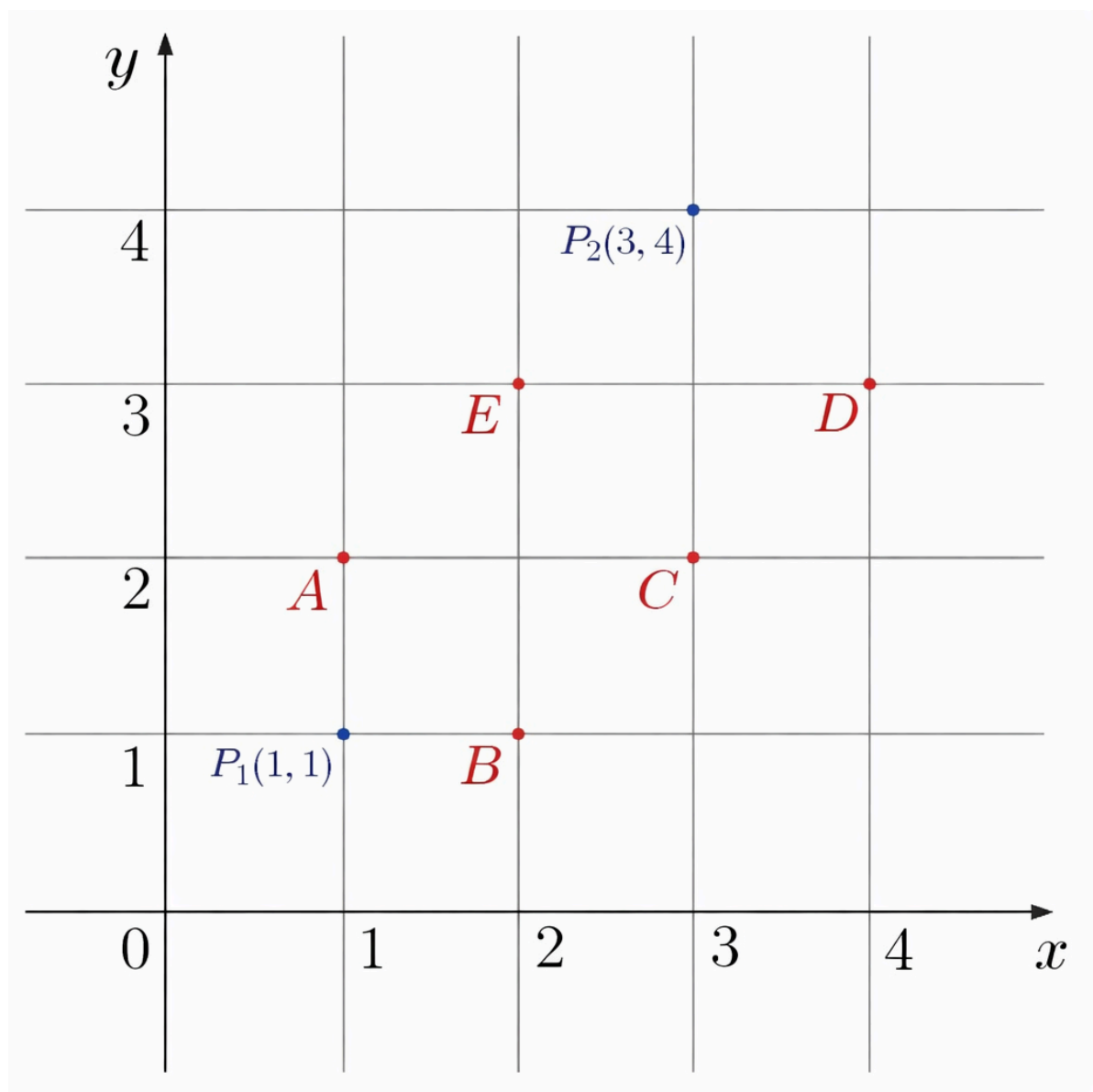
E 10

WORKED SOLUTION

Let there be m monkeys. The two descriptions give $N = 8m$ and $N = 10(m - 2)$. Thus $8m = 10m - 20$, so $m = 10$ and $N = 80$. Digit sum = 8.

QUESTION 23

Every point on the plane can be assigned with a coordinate that has two numbers. For example, the coordinate of point P_1 in the following picture is $(1, 1)$ and the coordinate of point P_2 is $(3, 4)$. Which of the following points has the coordinate $(2, 3)$?



A A

B B

C C

D D

E E

WORKED SOLUTION

Read the grid directly: the point at (2, 3) is labelled E.

M1-24

Mock Test 1 - Mathematics 1

Answer **D**

QUESTION 24

A pizza-shop offers a basic version of pizza with mozzarella and tomatoes. Two different toppings from the following list must be added: anchovies, artichokes, mushrooms, capers. Moreover, for each pizza three different sizes are available: small, medium, large. How many different types of pizza are available at all?

A 11

B 12

C 15

D 18

E 36

WORKED SOLUTION

Choose 2 toppings from 4: $\binom{4}{2} = 6$. For each, there are 3 sizes, so total pizzas $= 6 \cdot 3 = 18$.

M1-25

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 25

Bag A: 1 Snickers bar + 1 Butterfinger bar.

Bag B: 1 Butterfinger bar + 3 Reese's Peanut Butter Cups.

Bag C: 1 Snickers bar + 2 Reese's Peanut Butter Cups.

If each bag costs 50 cents, what is the price of
1 Snickers bar + 1 Butterfinger bar + 1 Reese's Peanut Butter Cup,
in cents?

A 40

B 45

C 50

D 55

E 60

WORKED SOLUTION

Let the prices be S , B , R . The bags give $S + B = 50$, $B + 3R = 50$, and $S + 2R = 50$. Solving gives $S = 30$, $B = 20$, $R = 10$, so $S + B + R = 60$.

M1-26

Mock Test 1 - Mathematics 1

Answer C

QUESTION 26

For how many positive integers x is

$$(x - 1)(x - 4)(x - 9)$$

negative?

A 2

B 3

C 4

D 5

E 6

WORKED SOLUTION

The roots are 1, 4 and 9. Testing the sign in the intervals shows the product is negative for $x < 1$ and $4 < x < 9$. The positive integers satisfying this are 5, 6, 7, 8, so there are 4 values.

M1-27

Mock Test 1 - Mathematics 1

Answer **C**

QUESTION 27

A gumball machine contains 9 red, 7 white, and 8 blue gumballs. The least number of gumballs a person must buy to be sure of getting four gumballs of the same color is

A 8

B 9

C 10

D 12

E 18

WORKED SOLUTION

The most gumballs possible without having four of one colour is $3 + 3 + 3 = 9$.
One more guarantees four of some colour, so the least number is 10.

Physics

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Physics module. Question wording, option order, answer and solution method are unchanged.

P-01

Mock Test 1 - Physics

Answer **F**

QUESTION 1

Which of the following statements are **FALSE**?

- A** Electromagnetic waves cause things to heat up.
- B** X-rays and gamma rays can knock electrons out of their orbits.
- C** Loud sounds can make objects vibrate.
- D** Wave power can be used to generate electricity.
- E** Since waves carry energy away, the source of a wave loses energy.
- F** The amplitude of a wave determines its mass.

WORKED SOLUTION

Waves are not objects with mass. Amplitude affects energy/intensity, not mass, so statement F is false.

QUESTION 2

A spacecraft is analysing a newly discovered exoplanet. A rock of unknown mass falls from rest on the planet from a height of 30 m, ignoring air resistance. Given that $g = 5.4 \text{ m s}^{-2}$ on the planet, calculate the speed of the rock when it hits the ground and the time it took to fall.

	Speed (ms^{-1})	Time (s)
A	18	3.3
B	18	3.1
C	12	3.3
D	10	3.7
E	9	2.3
F	1	0.3

WORKED SOLUTION

Use $v^2 = u^2 + 2as$ with $u = 0$: $v^2 = 2(5.4)(30) = 324$, so $v = 18 \text{ m s}^{-1}$.

Also $s = \frac{1}{2}at^2$, so $30 = 2.7t^2$ and $t \approx 3.3 \text{ s}$.

QUESTION 3

A canoe floating on the sea rises and falls 7 times in 49 seconds. The waves pass it at a speed of 5 m s^{-1} . How long are the waves?

A 12 m

B 22 m

C 25 m

D 35 m

E 57 m

F 75 m

WORKED SOLUTION

The period is $T = 49/7 = 7$ s. Since $v = f\lambda = \lambda/T$, $\lambda = vT = 5 \cdot 7 = 35$ m.

P-04

Mock Test 1 - Physics

Answer **E**

QUESTION 4

Miss Orrell lifts her 37.5 kg bike for a distance of 1.3 m in 5 s. The acceleration of free fall is 10 m s^{-2} . What is the average power that she develops?

A 9.8 W

B 12.9 W

C 57.9 W

D 79.5 W

E 97.5 W

F 98.0 W

WORKED SOLUTION

Work done = $mgh = 37.5 \cdot 10 \cdot 1.3 = 487.5 \text{ J}$. Power = $487.5 / 5 = 97.5 \text{ W}$.

P-05

Mock Test 1 - Physics

Answer **G**

QUESTION 5

A truck accelerates at 5.6 m s^{-2} from rest for 8 seconds. Calculate the final speed and the distance travelled in 8 seconds.

	Final Speed (ms^{-1})	Distance (m)
A	40.8	119.2
B	40.8	129.6
C	42.8	179.2
D	44.1	139.2
E	44.1	179.7
F	44.2	129.2
G	44.8	179.2
H	44.8	179.7

WORKED SOLUTION

From rest, $v = at = 5.6 \cdot 8 = 44.8 \text{ m s}^{-1}$. Distance

$$s = \frac{1}{2}at^2 = \frac{1}{2} \cdot 5.6 \cdot 64 = 179.2 \text{ m}.$$

QUESTION 6

Which of the following statements is true when a sky diver jumps out of a plane?

- A** The sky diver leaves the plane and will accelerate until the air resistance is greater than their weight.
- B** The sky diver leaves the plane and will accelerate until the air resistance is less than their weight.
- C** The sky diver leaves the plane and will accelerate until the air resistance equals their weight.
- D** The sky diver leaves the plane and will accelerate until the air resistance equals their weight squared.
- E** The sky diver will travel at a constant velocity after leaving the plane.

WORKED SOLUTION

Initially weight is greater than air resistance, so the skydiver accelerates. As speed rises, air resistance increases until it equals weight; then acceleration becomes zero.

QUESTION 7

A 100 g apple falls from rest on Isaac's head from a height of 20 m, ignoring air resistance. Calculate the apple's momentum before the point of impact. Take $g = 10 \text{ m s}^{-2}$.

A 0.1 kg m s^{-1}

B 0.2 kg m s^{-1}

C 1 kg m s^{-1}

D 2 kg m s^{-1}

E 10 kg m s^{-1}

F 20 kg m s^{-1}

WORKED SOLUTION

The apple mass is 0.1 kg . Its speed before impact is

$$v = \sqrt{2gh} = \sqrt{2 \cdot 10 \cdot 20} = 20 \text{ m s}^{-1}. \text{ Momentum}$$

$$p = mv = 0.1 \cdot 20 = 2 \text{ kg m s}^{-1}.$$

P-08

Mock Test 1 - Physics

Answer E

QUESTION 8

Which of the following do all electromagnetic waves have in common?

1. They can travel through a vacuum.
2. They can be reflected.
3. They are the same length.
4. They have the same amount of energy.
5. They can be polarised.

A 1, 2 and 3 only

B 1, 2, 3 and 4 only

C 4 and 5 only

D 3 and 4 only

E 1, 2 and 5 only

F 1 and 5 only

WORKED SOLUTION

All electromagnetic waves can travel through vacuum, reflect, and be polarised. They do not all have the same wavelength or energy.

P-09

Mock Test 1 - Physics

Answer **D**

QUESTION 9

A battery with an internal resistance of $0.8\ \Omega$ and e.m.f of $36\ \text{V}$ is used to power a drill with resistance $1\ \Omega$. What is the current in the circuit when the drill is connected to the power supply?

A 5 A

B 10 A

C 15 A

D 20 A

E 25 A

F 30 A

WORKED SOLUTION

The internal and drill resistances are in series, so $R_{\text{total}} = 0.8 + 1 = 1.8 \Omega$.

Current $I = V/R = 36/1.8 = 20 \text{ A}$.

P-10

Mock Test 1 - Physics

Answer D

QUESTION 10

Officer Bailey throws a 20 g dart at a speed of 100 m s^{-1} . It strikes the dartboard and is brought to rest in 10 milliseconds. Calculate the average force exerted on the dart by the dartboard.

A 0.2 N

B 2 N

C 20 N

D 200 N

E 2000 N

F 20000 N

WORKED SOLUTION

Mass = 0.020 kg, change in velocity = 100 m s^{-1} , and time = 0.010 s.

Average force $F = \Delta p / \Delta t = 0.020 \cdot 100 / 0.010 = 200 \text{ N}$.

P-11

Mock Test 1 - Physics

Answer **F**

QUESTION 11

Professor Huang lifts a 50 kg bag through a distance of 0.7 m in 3 s. What average power does she develop to 3 significant figures? Take $g = 10 \text{ m s}^{-2}$.

A 112 W

B 113 W

C 114 W

D 115 W

E 116 W

F 117 W

WORKED SOLUTION

Work done = $mgh = 50 \cdot 10 \cdot 0.7 = 350 \text{ J}$. Power = $350/3 = 116.7 \text{ W}$, which is 117 W to 3 significant figures.

P-12

Mock Test 1 - Physics

Answer **B**

QUESTION 12

An electric scooter is travelling at a speed of 30 m s^{-1} . The motor provides a driving force of 300 N in the direction of motion. Assuming all electrical power is converted into mechanical power, and given that the motor runs at 200 V , calculate the current in the scooter.

A 4.5 A

B 45 A

C 450 A

D 4500 A

E 45000 A

F More information needed.

WORKED SOLUTION

Use the motor driving force for power: $P = Fv = 300 \cdot 30 = 9000 \text{ W}$. Since $P = VI$, $I = 9000/200 = 45 \text{ A}$.

QUESTION 13

Which of the following statements about the physical definition of work are correct?

1. Work done = $\frac{\text{Force}}{\text{distance}}$
2. The unit of work is equivalent to kg m s^{-2} .
3. Work is defined as a force causing displacement of the body upon which it acts.

A Only 1

B Only 2

C Only 3

D 1 and 2

E 2 and 3

F 1 and 3

WORKED SOLUTION

Work is force times displacement in the direction of the force, not force divided by distance. Its unit is the joule, $\text{kg m}^2\text{s}^{-2}$, not kg m s^{-2} . Only statement 3 is correct.

QUESTION 14

Which of the following statements about kinetic energy are correct?

1. It is defined as $E_k = \frac{mv^2}{2}$.
2. The unit of kinetic energy is equivalent to $\text{Pa} \times \text{m}^3$.
3. Kinetic energy is equal to the amount of energy needed to decelerate the body in question from its current speed.

A Only 1

B Only 2

C Only 3

D 1 and 2

E 2 and 3

F 1 and 3

G 1, 2 and 3

WORKED SOLUTION

$E_k = \frac{1}{2}mv^2$. Also $\text{Pa} \cdot \text{m}^3 = (\text{N}/\text{m}^2)\text{m}^3 = \text{N m} = \text{J}$. Kinetic energy is the work needed to bring the body to rest, so all three are correct.

P-15

Mock Test 1 - Physics

Answer D

QUESTION 15

In relation to radiation, which of the following statements is **FALSE**?

A Radiation is the emission of energy in the form of waves or particles.

B Radiation can be either ionizing or non-ionizing.

C Gamma radiation has very high energy.

D Alpha radiation is of higher energy than beta radiation.

E X-rays are an example of wave radiation.

WORKED SOLUTION

Radiation can be waves or particles, and gamma rays and X-rays are high-energy electromagnetic radiation. The statement that alpha radiation is always of higher energy than beta radiation is not a reliable general definition, so D is false.

P-16

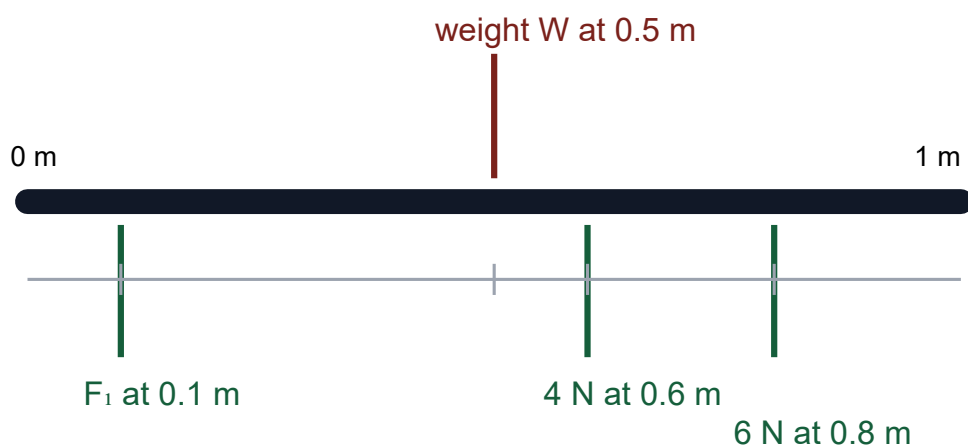
Mock Test 1 - Physics

Answer E

QUESTION 16

A uniform rod of length 1 m is balanced on 3 supports so that it is in equilibrium. The position of the first support is at $x_1 = 0.1$ m, where x is the distance along the rod. The second support is at $x_2 = 0.6$ m and applies a force, F_2 , of 4 N to the rod. The third support is at $x_3 = 0.8$ m and applies a force, F_3 , of 6 N to the rod.

What is the force applied by the first support, F_1 ?



A 4.5 N

B 5.0 N

C 5.3 N

D 6.7 N

E 5.5 N

F 11.0 N

G 10.5 N

H 7.3 N

WORKED SOLUTION

Let the rod weight be W . Force balance gives $W = F_1 + 4 + 6 = F_1 + 10$.

Taking moments about the left end: $0.1F_1 + 0.6(4) + 0.8(6) = 0.5W$. Hence $0.1F_1 + 7.2 = 0.5(F_1 + 10)$, so $F_1 = 5.5$ N.

P-17

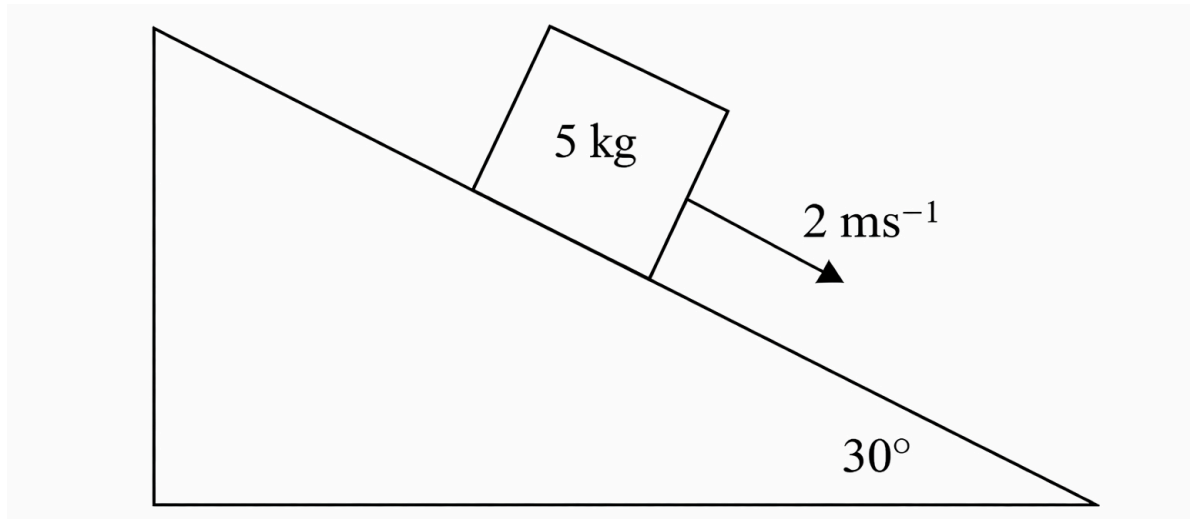
Mock Test 1 - Physics

Answer B

QUESTION 17

A block of mass 5 kg slides down a rough slope angled at 30° to the horizontal. The block moves at a constant velocity of 2 m s^{-1} . (Take $g = 10 \text{ m s}^{-2}$.)

What is the frictional force on the block?



A 14 N

B 25 N

C $\frac{50}{\sqrt{2}} \text{ N}$

D $50\sqrt{2} \text{ N}$

E $25\sqrt{3} \text{ N}$

F 55 N

G $50\sqrt{3} \text{ N}$

H 50 N

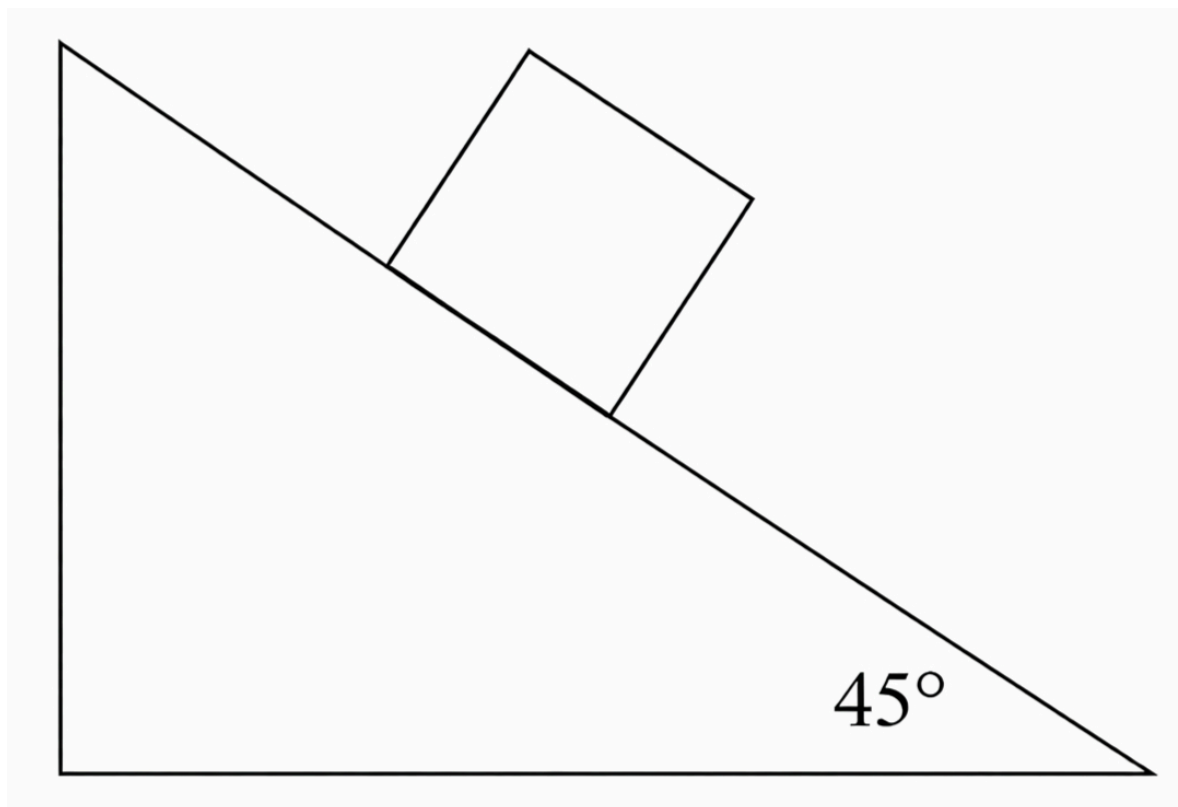
WORKED SOLUTION

Constant velocity means zero resultant force along the slope. The downslope component of weight is $mg \sin 30^\circ = 5 \cdot 10 \cdot \frac{1}{2} = 25 \text{ N}$, so friction is 25 N upslope.

QUESTION 18

A block slides down a smooth slope angled at 45° to the horizontal. The block is initially stationary. (Take $g = 10 \text{ m s}^{-2}$.)

What is the velocity of the block after 10 s?



A 50 m s^{-1}

B 40 m s^{-1}

C $\frac{100}{\sqrt{2}} \text{ m s}^{-1}$

D $\frac{200}{\sqrt{3}} \text{ m s}^{-1}$

E $\frac{100}{\sqrt{3}} \text{ m s}^{-1}$

F 75 m s^{-1}

G $\frac{200}{3} \text{ m s}^{-1}$

H $\frac{200}{\sqrt{2}} \text{ m s}^{-1}$

WORKED SOLUTION

On a smooth slope, acceleration down the plane is $g \sin 45^\circ = 10/\sqrt{2}$. Starting from rest for 10 seconds gives $v = at = 100/\sqrt{2} \text{ m s}^{-1}$.

P-19

Mock Test 1 - Physics

Answer G

QUESTION 19

An object of mass 10 kg falls from a great height. If the force due to air resistance can be modelled as $F_a = 0.5u \text{ N}$ where u is the velocity of the free-falling object, what is the terminal velocity of the object? (Take $g = 10 \text{ N kg}^{-1}$.)

A 700 m s^{-1}

B 150 m s^{-1}

C 50 m s^{-1}

D 100 m s^{-1}

E 800 m s^{-1}

F 400 m s^{-1}

G 200 m s^{-1}

H 500 m s^{-1}

WORKED SOLUTION

At terminal velocity, air resistance equals weight: $0.5u = mg = 10 \cdot 10 = 100$.

Therefore $u = 200 \text{ m s}^{-1}$.

P-20

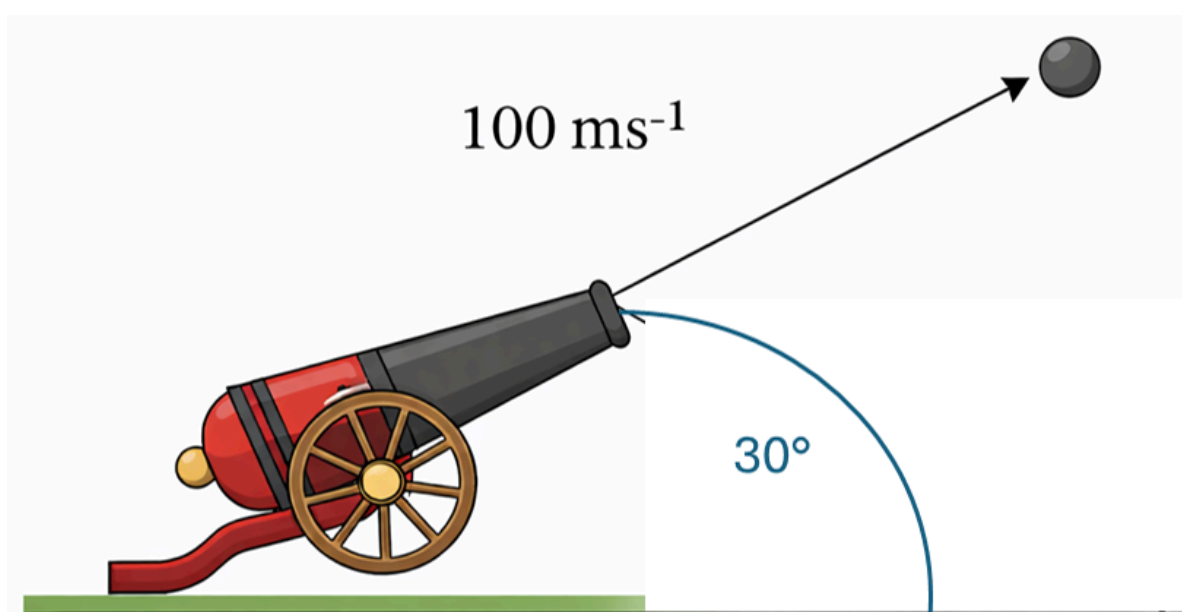
Mock Test 1 - Physics

Answer **D**

QUESTION 20

A cannonball is fired from ground level at 30° to the horizontal with an initial velocity of 100 m s^{-1} . The ground is flat and air resistance is negligible. (Take $g = 10 \text{ N kg}^{-1}$.)

How far does the cannonball travel horizontally before it hits the ground?



A 250 m

B $250\sqrt{2} \text{ m}$

C 300 m

D $500\sqrt{3}$ m

E 500 m

F $\frac{500}{\sqrt{2}}$ m

G $600\sqrt{3}$ m

H $600\sqrt{2}$ m

WORKED SOLUTION

Use projectile range $R = \frac{u^2 \sin 2\theta}{g}$. Thus

$$R = \frac{100^2 \sin 60^\circ}{10} = 1000 \cdot \frac{\sqrt{3}}{2} = 500\sqrt{3} \text{ m.}$$

P-21

Mock Test 1 - Physics

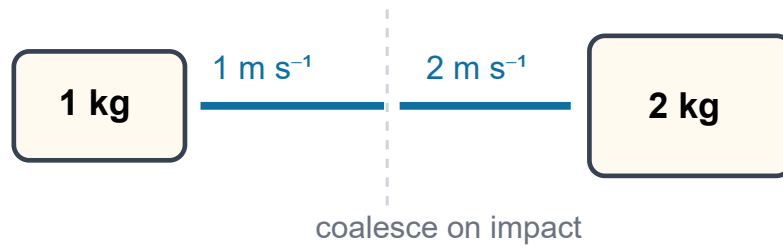
Answer E

QUESTION 21

Two blocks, one of mass 1 kg and one of mass 2 kg, move directly towards each other and collide inelastically. Before collision, their velocities have magnitudes of 1 m s^{-1} and 2 m s^{-1} respectively.

Assuming the two blocks coalesce on impact, what fraction of the kinetic energy is lost (e.g. to heat)?

Before collision



A 1

B 0

C $\frac{1}{3}$

D $\frac{1}{2}$

E $\frac{2}{3}$

F $\frac{1}{4}$

G $\frac{3}{4}$

H $\frac{2}{5}$

WORKED SOLUTION

Initial momentum is $1(1) - 2(2) = -3$, so after coalescing,
 $v = -3/(1 + 2) = -1 \text{ m s}^{-1}$. Initial KE is $\frac{1}{2}(1)(1^2) + \frac{1}{2}(2)(2^2) = 4.5$. Final
KE is $\frac{1}{2}(3)(1^2) = 1.5$. Fraction lost = $(4.5 - 1.5)/4.5 = 2/3$.

QUESTION 22

An object of mass 20 kg falls 100 m and experiences an average resistive force of 40 N due to air resistance. If the object is initially at rest, what is its final velocity? (Take $g = 10 \text{ N kg}^{-1}$.)

A 60 m s^{-1}

B 420 m s^{-1}

C 360 m s^{-1}

D 200 m s^{-1}

E 80 m s^{-1}

F 500 m s^{-1}

G 40 m s^{-1}

H 800 m s^{-1}

WORKED SOLUTION

Energy lost to air resistance is $40 \cdot 100 = 4000 \text{ J}$. Gravitational energy lost is $mgh = 20 \cdot 10 \cdot 100 = 20000 \text{ J}$. Final KE = $16000 \text{ J} = \frac{1}{2}(20)v^2$, so $v = 40 \text{ m s}^{-1}$.

QUESTION 23

A velocity profile is given by

$$v = \frac{(5t + 3)^2}{t^{\frac{1}{2}}}$$

(SI units). Find the total distance travelled from $t = 1$ to $t = 4$ s.

A 490 m

B 955 m

C 1082 m

D 683 m

E 575 m

F 23 m

G 1820 m

H 468 m

WORKED SOLUTION

Rewrite $v = (5t + 3)^2 t^{-1/2} = 25t^{3/2} + 30t^{1/2} + 9t^{-1/2}$. Integrating gives $10t^{5/2} + 20t^{3/2} + 18t^{1/2}$. From 1 to 4:
 $(320 + 160 + 36) - (10 + 20 + 18) = 468$ m.

QUESTION 24

A spring is used to lift 20 cm^3 of loosely packed sand in a small bucket. The spring constant, k , is 4 N m^{-1} and the sand causes the spring to extend by 10 cm . What is the density of the sand at this packing fraction? (Take $g = 10 \text{ N kg}^{-1}$.)

A 3200 kg m^{-3}

B 32000 kg m^{-3}

C 2000 kg m^{-3}

D 1000 kg m^{-3}

E 20000 kg m^{-3}

F 2500 kg m^{-3}

G 4500 kg m^{-3}

H 800 kg m^{-3}

WORKED SOLUTION

The spring force is $F = kx = 4 \cdot 0.10 = 0.4 \text{ N}$, so mass $m = F / g = 0.04 \text{ kg}$.

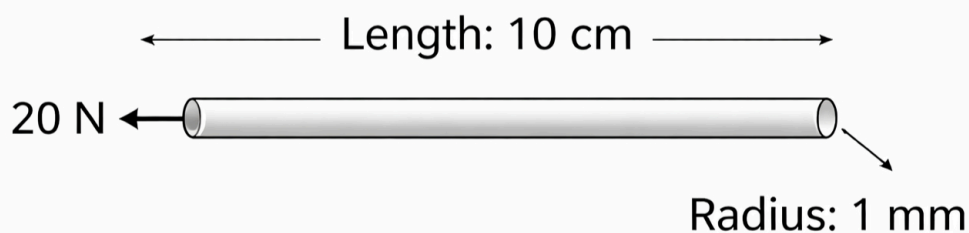
Volume $20 \text{ cm}^3 = 2.0 \times 10^{-5} \text{ m}^3$. Density

$= 0.04 / (2.0 \times 10^{-5}) = 2000 \text{ kg m}^{-3}$.

QUESTION 25

A cylindrical wire of radius 1 mm, and length 10 cm is extended by applying a tension of 20 N to it, resulting in an elastic extension of 2 mm.

What is the Young's modulus of the material?



A $2\pi \text{ GPa}$

B $\frac{2}{\pi} \text{ GPa}$

C $\pi \text{ GPa}$

D $\frac{1}{\pi} \text{ GPa}$

E $20\pi \text{ MPa}$

F $\frac{20}{\pi} \text{ MPa}$

G $10\pi \text{ GPa}$

H $\frac{10}{\pi} \text{ GPa}$

WORKED SOLUTION

Young's modulus $E = \frac{FL}{A\Delta L}$. With $F = 20$, $L = 0.10$, $A = \pi(10^{-3})^2$, and $\Delta L = 2 \times 10^{-3}$, $E = \frac{1}{\pi} \times 10^9 \text{ Pa} = \frac{1}{\pi} \text{ GPa}$.

P-26

Mock Test 1 - Physics

Answer C

QUESTION 26

A uranium-235 nucleus can undergo fission to produce two smaller nuclei. Which of the diagrams, if any, could represent this process?

Diagram 1

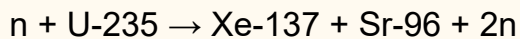


Diagram 2

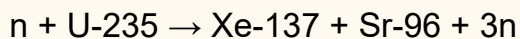
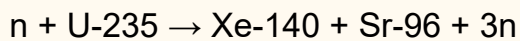


Diagram 3



A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

WORKED SOLUTION

Check conservation of nucleon number and proton number. Diagram 2 conserves both: $1 + 235 = 137 + 96 + 3$ and $92 = 54 + 38$. Diagrams 1 and 3 do not conserve mass number as drawn.

P-27

Mock Test 1 - Physics

Answer **B**

QUESTION 27

A wave travels at 30 m in 10 seconds and has a frequency, f , of 2 Hz. How many complete wave cycles occur if the wave travels 120 m?

A 100

B 80

C 50

D 40

E 120

F 20

G 30

H 240

WORKED SOLUTION

The wave speed is $30/10 = 3 \text{ m s}^{-1}$. Wavelength $\lambda = v/f = 3/2 = 1.5 \text{ m}$.
Over 120 m, the number of cycles is $120/1.5 = 80$.

Chemistry

27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Chemistry Easy source set. Question wording, option order, answer and solution method are unchanged.

C-01

Periodic Table and electron configuration

Answer B**QUESTION 1**

Four atoms have the following electron configurations.

atom	electron configuration
P	2,8,8
Q	2,8,1
R	2,8,7
S	2,8,8,1

Which statement is correct?

- A** P is a Period 2 noble gas.
- B** Q and S are in Group 1, and S is expected to react more vigorously with water than Q.
- C** R normally forms R^{2-} ions.
- D** P has one electron in its outer shell.
- E** Q is less metallic than R.

WORKED SOLUTION

Q has electron configuration 2,8,1 and S has electron configuration 2,8,8,1, so both are in Group 1.

Group 1 reactivity increases down the group. S has one more occupied electron shell than Q, so S is below Q and is expected to react more vigorously with water.

P has three occupied shells and a full outer shell, so it is a Period 3 noble gas. R has seven outer-shell electrons and normally forms a 1^- ion.

Therefore the correct answer is **B**.

C-02

Formulae of ionic compounds

Answer **D**

QUESTION 2

Aluminium ions are Al^{3+} and sulfate ions are SO_4^{2-} .

Which row gives the formula of aluminium sulfate and the number of sulfate ions in one formula unit?

option	formula	sulfate ions
A	AlSO_4	1
B	$\text{Al}(\text{SO}_4)_2$	2
C	Al_2SO_4	1
D	$\text{Al}_2(\text{SO}_4)_3$	3
E	$\text{Al}_3(\text{SO}_4)_2$	2

A A

B B

C C

D D

E E

WORKED SOLUTION

The smallest common magnitude of charge is 6:

$$2 \times (+3) = +6, \quad 3 \times (-2) = -6.$$

Therefore two aluminium ions combine with three sulfate ions:



There are three sulfate ions in one formula unit, so the correct row is **D**.

C-03

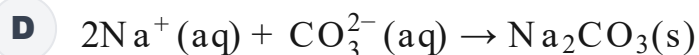
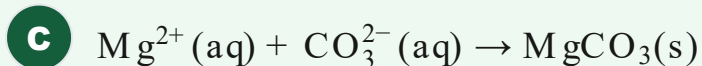
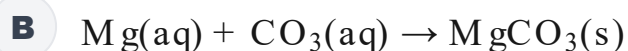
Ionic equations

Answer C

QUESTION 3

Aqueous magnesium chloride is mixed with aqueous sodium carbonate. A white precipitate of magnesium carbonate forms.

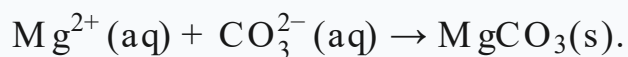
Which is the ionic equation for this reaction?



WORKED SOLUTION

The sodium and chloride ions remain dissolved and are spectator ions.

The ions that actually form the precipitate are magnesium ions and carbonate ions:



Therefore the answer is **C**.

C-04

Moles and particles

Answer **D**

QUESTION 4

How many moles of helium atoms contain the same total number of **atoms** as 0.20 mol of oxygen molecules, O_2 ?

A 0.10 mol

B 0.20 mol

C 0.25 mol

D 0.40 mol

E 0.80 mol

WORKED SOLUTION

Each oxygen molecule contains two oxygen atoms. Therefore 0.20 mol of O_2 contains

$$2(0.20) = 0.40 \text{ mol}$$

of atoms.

Helium is monatomic, so 0.40 mol of helium contains the same number of atoms.

The answer is **D**.

QUESTION 5

A 2.50 g impure sample of calcium carbonate contains impurities that do not react with hydrochloric acid.

The sample is reacted with excess hydrochloric acid and produces 0.480 dm^3 of carbon dioxide at room temperature and pressure.

At rtp,

$$1 \text{ mol of gas} = 24.0 \text{ dm}^3.$$

$$A_r : \quad \text{Ca} = 40, \text{ C} = 12, \text{ O} = 16$$

What is the percentage purity of the calcium carbonate?

A 40%

B 60%

C 75%

D 80%

E 90%

WORKED SOLUTION

The amount of carbon dioxide formed is

$$n(\text{CO}_2) = \frac{0.480}{24.0} = 0.0200 \text{ mol.}$$

From



the sample contained 0.0200 mol of calcium carbonate.

$$M_r(\text{CaCO}_3) = 40 + 12 + 3(16) = 100.$$

So the mass of pure calcium carbonate was

$$0.0200 \times 100 = 2.00 \text{ g.}$$

Hence

$$\text{purity} = \frac{2.00}{2.50} \times 100 = 80\%.$$

The answer is **D**.

C-06

Kinetic particle theory

Answer **D**

QUESTION 6

A pure substance is heated. During melting, energy continues to be supplied but its temperature remains constant until all of the solid has melted.

Which statement best explains this?

- A** The particles stop moving while the substance melts.
- B** The supplied energy breaks the atoms into smaller particles.
- C** The supplied energy increases the average kinetic energy of the particles, but the temperature does not change.
- D** The supplied energy is used in overcoming forces between particles rather than increasing their average kinetic energy.
- E** Particles in the solid and liquid have exactly the same arrangement.

WORKED SOLUTION

Temperature is related to the average kinetic energy of the particles.

During melting the temperature is constant, so the average kinetic energy is not increasing. The supplied energy is instead used to overcome forces holding particles in the solid structure.

Therefore the correct answer is **D**.

C-07

Metallic bonding

Answer **F**

QUESTION 7

Consider the following statements about a solid metal.

1. Some electrons are delocalised and can move through the structure.
2. Electrical conduction occurs because positive metal ions move through the solid.
3. Layers of metal ions can move relative to one another while metallic bonding is maintained.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: metals contain delocalised electrons that can move through the structure.

Statement 2 is false: the positive metal ions are fixed in the solid lattice; electrical conduction is due to mobile delocalised electrons.

Statement 3 is correct: layers can slide while attraction between positive ions and delocalised electrons is maintained.

So statements 1 and 3 only are correct. The answer is **F**.

C-08

Giant structures

Answer C

QUESTION 8

Which substance has a very high melting point, does not conduct electricity when solid, and does not become an electrical conductor simply on melting?

A sodium chloride

B graphite

C silicon dioxide

D copper

E methane

WORKED SOLUTION

Silicon dioxide has a giant covalent structure. Many strong covalent bonds must be overcome to melt it, so its melting point is very high.

It does not have mobile ions or delocalised electrons available for electrical conduction, whether solid or molten.

Sodium chloride is different because its ions become mobile when molten.

Therefore the answer is **C**.

C-09

Group 1 trends

Answer **B**

QUESTION 9

Which row correctly shows the trends from lithium to sodium to potassium?

option	reactivity with water	melting point
A	$\text{Li} > \text{Na} > \text{K}$	$\text{Li} > \text{Na} > \text{K}$
B	$\text{K} > \text{Na} > \text{Li}$	$\text{Li} > \text{Na} > \text{K}$
C	$\text{K} > \text{Na} > \text{Li}$	$\text{K} > \text{Na} > \text{Li}$
D	$\text{Li} > \text{Na} > \text{K}$	$\text{K} > \text{Na} > \text{Li}$
E	$\text{Na} > \text{K} > \text{Li}$	$\text{Li} > \text{K} > \text{Na}$

A A

B B

C C

D D

E E

WORKED SOLUTION

Group 1 metals become more reactive down the group:



Their melting points decrease down the group:



Only row **B** gives both trends correctly.

C-10

Air and drinking water

Answer **H**

QUESTION 10

Consider these statements.

1. Dry air contains approximately 78% nitrogen.
2. Chlorine may be added during drinking-water treatment to kill harmful microorganisms.
3. Fluoride ions may be added to drinking water to help reduce tooth decay.
4. Components of air can be separated by fractional distillation after the air has been liquefied.

Which statements are correct?

- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 4 only
- D** 1, 2 and 3 only
- E** 1, 2 and 4 only
- F** 1, 3 and 4 only
- G** 2, 3 and 4 only
- H** 1, 2, 3 and 4

WORKED SOLUTION

All four statements are correct.

Dry air is about 78% nitrogen. Chlorine is used for disinfection in water treatment, fluoride ions may be added to help reduce tooth decay, and the major gases in air can be separated by fractional distillation after liquefaction.

Therefore the answer is **H**.

C-11

Separation techniques

Answer **C**

QUESTION 11

A mixture contains:

- insoluble sand,
- hexane,
- water containing dissolved sodium chloride.

Hexane and water are immiscible.

Which sequence is suitable for first removing the sand, then isolating the hexane, and finally obtaining sodium chloride crystals?

A separating funnel → filtration → evaporation

B filtration → crystallisation → separating funnel

C filtration → separating funnel → crystallisation

D crystallisation → filtration → separating funnel

E centrifugation → chromatography → filtration

WORKED SOLUTION

First remove the insoluble sand by filtration.

The filtrate contains two immiscible liquid layers, hexane and aqueous sodium chloride, so a separating funnel can isolate the hexane.

Finally, sodium chloride crystals can be obtained from the aqueous layer by crystallisation.

The correct sequence is therefore **filtration** → **separating funnel** → **crystallisation**, answer **C**.

C-12

Obtaining a pure solvent

Answer **A**

QUESTION 12

A mixture contains water, dissolved sodium chloride, and a small amount of insoluble charcoal.

Which procedure is most suitable for obtaining a sample of **pure water** from the mixture?

- A** Filter, then use simple distillation on the filtrate.
- B** Filter, then evaporate the filtrate to dryness.
- C** Use a separating funnel, then crystallise.
- D** Use fractional distillation without first filtering.
- E** Use paper chromatography.

WORKED SOLUTION

The insoluble charcoal should first be removed by filtration.

The filtrate is salt water. Sodium chloride is non-volatile, while water can be vapourised and condensed, so simple distillation gives pure water as the distillate.

Therefore the answer is **A**.

QUESTION 13

A strong acid has pH 2.

It is diluted with water until the total volume is 100 times its original volume. Assume the acid remains fully ionised.

What is its new pH?

A 1

B 2

C 3

D 4

E 6

WORKED SOLUTION

A 100-fold dilution reduces the hydrogen-ion concentration by a factor of

$$100 = 10^2.$$

Each factor of 10 reduction in $[H^+]$ increases pH by 1, so the pH increases by 2:

$$2 + 2 = 4.$$

Therefore the answer is **D**.

QUESTION 14

20.0 cm³ of 0.30 mol dm⁻³ sulfuric acid is completely neutralised by 0.50 mol dm⁻³ sodium hydroxide.

Assume one mole of sulfuric acid supplies two moles of H⁺.

What volume of sodium hydroxide is required?

A 12.0 cm³

B 18.0 cm³

C 24.0 cm³

D 30.0 cm³

E 48.0 cm³

WORKED SOLUTION

The amount of sulfuric acid is

$$n(\text{H}_2\text{SO}_4) = 0.30 \times 0.0200 = 0.00600 \text{ mol.}$$

Each mole supplies two moles of H⁺, so 0.0120 mol of OH⁻ is needed.

Therefore

$$V(\text{NaOH}) = \frac{0.0120}{0.50} = 0.0240 \text{ dm}^3 = 24.0 \text{ cm}^3.$$

The answer is **C**.

QUESTION 15

A student wants to prepare pure, dry crystals of copper(II) sulfate using dilute sulfuric acid and insoluble copper(II) oxide.

Which method is best?

- A** Add exactly one spatula of copper oxide to cold acid and evaporate immediately to dryness.
- B** Warm the acid, add copper oxide until some remains unreacted, filter, partially evaporate the filtrate, cool it, then filter and dry the crystals.
- C** Add an indicator to the acid and titrate it with solid copper oxide.
- D** Dissolve the copper oxide in water, filter, then add sulfuric acid.
- E** Mix the reagents and collect the product by simple distillation.

WORKED SOLUTION

Copper(II) oxide is insoluble. Adding it in excess ensures that all the sulfuric acid has reacted:



Unreacted copper oxide is removed by filtration. The filtrate is then partially evaporated and cooled to crystallise copper(II) sulfate, after which the crystals are filtered and dried.

Therefore the answer is **B**.

QUESTION 16

Magnesium carbonate reacts with excess dilute hydrochloric acid.



Which measurement could be used to follow the rate of the reaction **without collecting the carbon dioxide gas**?

- A** Measure the total mass of the reaction vessel and contents at regular time intervals.
- B** Measure only the final pH.
- C** Measure the original volume of hydrochloric acid.
- D** Measure the mass of magnesium chloride before the reaction begins.
- E** Measure only the room temperature.

WORKED SOLUTION

Carbon dioxide leaves the reaction vessel as the reaction proceeds, so the total mass of the vessel and contents decreases.

Recording the mass at regular time intervals therefore provides a measure of the reaction progress and hence its rate.

The answer is **A**.

QUESTION 17

Three experiments use the same mass of calcium carbonate and excess hydrochloric acid at the same temperature.

experiment	calcium carbonate	acid concentration
1	chips	1.0 mol dm^{-3}
2	powder	1.0 mol dm^{-3}
3	chips	2.0 mol dm^{-3}

The same volume of acid is used each time.

Consider the statements.

1. Experiment 2 should reach completion sooner than experiment 1.
2. Experiment 3 should produce twice the final volume of carbon dioxide as experiment 1.
3. Experiments 2 and 3 should both have greater initial rates than experiment 1.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Powder has a larger surface area than chips, so experiment 2 has a greater rate and should reach completion sooner than experiment 1. Statement 1 is true.

Experiment 3 has a higher acid concentration, so it also has a greater initial rate. Statement 3 is true.

However, the same mass of calcium carbonate is used and the acid is in excess in every experiment. Calcium carbonate therefore determines the total amount of carbon dioxide, so the final gas volume is the same. Statement 2 is false.

Thus statements 1 and 3 only are correct: **F**.

C-18

Electrolysis and half-equations

Answer **B**

QUESTION 18

Molten magnesium bromide is electrolysed using inert electrodes.

Which row correctly explains why **direct current** is used and gives the half-equation at the **anode**?

option	reason for using dc	anode half-equation
A	keeps electrode polarity fixed	$\text{Mg}^{2+} + 2\text{e}^{-} \rightarrow \text{Mg}$
B	keeps electrode polarity fixed	$2\text{Br}^{-} \rightarrow \text{Br}_2 + 2\text{e}^{-}$
C	continually reverses electrode polarity	$2\text{Br}^{-} \rightarrow \text{Br}_2 + 2\text{e}^{-}$
D	continually reverses electrode polarity	$\text{Mg}^{2+} + 2\text{e}^{-} \rightarrow \text{Mg}$
E	prevents ions moving	$\text{Br}_2 + 2\text{e}^{-} \rightarrow 2\text{Br}^{-}$

A A

B B

C C

D D

E E

WORKED SOLUTION

Direct current keeps each electrode at a fixed polarity. With alternating current the electrode polarity would repeatedly reverse.

At the positive electrode, bromide ions are oxidised:



Therefore row **B** is correct.

C-19

Hydrocarbon properties

Answer **C**

QUESTION 19

As the carbon-chain length of straight-chain alkanes increases from C_4 to C_8 , which combination of trends is expected?

option	boiling point	viscosity	flammability
A	decreases	decreases	decreases
B	increases	decreases	increases
C	increases	increases	decreases
D	decreases	increases	decreases
E	increases	decreases	decreases

A A

B B

C C

D D

E E

WORKED SOLUTION

As chain length increases, intermolecular attractions become stronger.

Therefore boiling point and viscosity increase. Larger hydrocarbons are less volatile and generally less readily flammable.

The correct combination is **increase, increase, decrease**, so the answer is **C**.

C-20

Structural isomerism

Answer **B**

QUESTION 20

Which pair consists of structural isomers?

A CH_4 and C_2H_6

B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$

C $\text{CH}_2=\text{CH}_2$ and CH_3CH_3

D $\text{CH}_3\text{CH}_2\text{OH}$ and CH_3COOH

E $\text{CH}_3\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

WORKED SOLUTION

Structural isomers have the same molecular formula but different structural formulae.

Both butane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$, and methylpropane, $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$, have molecular formula C_4H_{10} , but the atoms are connected differently.

Therefore the answer is **B**.

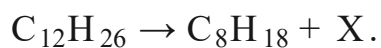
C-21

Cracking

Answer **C**

QUESTION 21

A hydrocarbon undergoes cracking:



What is X?

A C_2H_4

B C_3H_6

C C_4H_8

D C_4H_{10}

E C_5H_{10}

WORKED SOLUTION

Balance the carbon atoms:

$$12 = 8 + 4,$$

so X contains four carbon atoms.

Balance the hydrogen atoms:

$$26 = 18 + 8.$$

Therefore $X = C_4H_8$, which is consistent with cracking producing an alkene.

The answer is **C**.

C-22

Reactions of alkenes

Answer **D**

QUESTION 22

Ethene is shaken with bromine water.

Which row correctly gives the observation and the organic product?

option	observation	organic product
A	bromine colour remains	C_2H_6
B	bromine water decolourises	C_2H_6
C	bromine water decolourises	C_2H_5Br
D	bromine water decolourises	$C_2H_4Br_2$
E	solution turns blue	$C_2H_4Br_2$

A A

B B

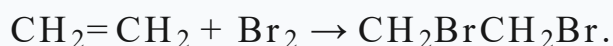
C C

D D

E E

WORKED SOLUTION

Bromine adds across the carbon-carbon double bond in ethene:



The bromine-water colour disappears and the product has molecular formula $C_2H_4Br_2$.

Therefore row **D** is correct.

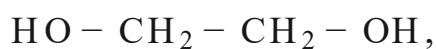
C-23

Condensation polymers

Answer **B**

QUESTION 23

A diol,



reacts with a dicarboxylic acid,



Which row correctly identifies the type of polymer formed and the small molecule eliminated during polymerisation?

- A** polyester; hydrogen
- B** polyester; water
- C** polyamide; water
- D** addition polymer; no small molecule
- E** polyamide; carbon dioxide

WORKED SOLUTION

A diol reacting with a dicarboxylic acid forms ester links, $-COO-$, between monomer units.

The polymer is therefore a polyester. Each condensation step eliminates a molecule of water.

The answer is **B**.

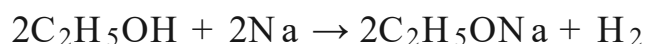
C-24

Alcohols

Answer **C**

QUESTION 24

Ethanol reacts with excess sodium.



What volume of hydrogen gas at rtp is formed when 4.60 g of ethanol reacts completely?

$$M_r(\text{C}_2\text{H}_5\text{OH}) = 46$$

and

$$1 \text{ mol gas} = 24.0 \text{ dm}^3.$$

A 0.60 dm³

B 0.80 dm³

C 1.20 dm³

D 2.40 dm³

E 4.80 dm³

WORKED SOLUTION

The amount of ethanol is

$$n = \frac{4.60}{46} = 0.100 \text{ mol.}$$

From the equation, two moles of ethanol produce one mole of hydrogen, so

$$n(\text{H}_2) = 0.0500 \text{ mol.}$$

At rtp,

$$V = 0.0500 \times 24.0 = 1.20 \text{ dm}^3.$$

The answer is **C**.

C-25

Carboxylic acids

Answer **C**

QUESTION 25

Ethanoic acid reacts completely with calcium carbonate.

Which pair gives the salt formed and the gas released?

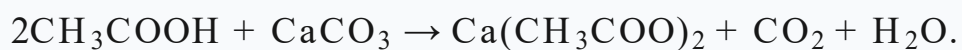
- A** $\text{CaCO}_3; \text{H}_2$
- B** $\text{Ca}(\text{CH}_3\text{COO}); \text{CO}_2$
- C** $\text{Ca}(\text{CH}_3\text{COO})_2; \text{CO}_2$
- D** $\text{Ca}(\text{CH}_3\text{COO})_2; \text{H}_2$
- E** $\text{CH}_3\text{COOCa}; \text{O}_2$

WORKED SOLUTION

An acid reacting with a carbonate produces a salt, carbon dioxide and water.

The ethanoate ion is CH_3COO^- and calcium is Ca^{2+} , so the salt is $\text{Ca}(\text{CH}_3\text{COO})_2$.

The balanced reaction is



Therefore the answer is **C**.

QUESTION 26

Consider these statements.

1. A metal that is more reactive than carbon cannot normally be extracted from its oxide by reduction with carbon.
2. Compounds of transition metals are often coloured.
3. Transition metals form ions in only one oxidation state.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: metals above carbon in the reactivity series cannot normally be extracted from their oxides using carbon reduction.

Statement 2 is correct: coloured compounds are a common characteristic of transition metals.

Statement 3 is false: transition metals commonly form ions in more than one oxidation state.

Therefore statements 1 and 2 only are correct, so the answer is **E**.

C-27

Chemical tests

Answer **C**

QUESTION 27

An unknown aqueous salt gives:

- a **crimson-red flame**;
- a **white precipitate** when aqueous barium chloride is added in the presence of dilute hydrochloric acid.

Which salt is the most likely identity of the unknown?

A LiCl

B Na₂SO₄

C Li₂SO₄

D K₂SO₄

E CuSO₄

F LiNO₃

WORKED SOLUTION

A crimson-red flame identifies lithium ions, Li⁺.

A white precipitate with acidified barium chloride indicates sulfate ions, SO₄²⁻, because barium sulfate is insoluble and white.

The salt must therefore be Li₂SO₄.

The answer is **C**.

Compilation record: Level 0 uses Engineering Mock Test 1 for Mathematics/Physics and the corresponding Easy Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.